

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION WITH OPENCV

COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

Assessment Tool Weightage: 40%

QUESTIONS

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.
2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.
3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.
4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.
5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.
6. Explain how image formation models help in improving image quality in applications such as autonomous driving.
7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.
8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.
9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.
10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

11. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.
12. A digital image appears distorted due to improper image formation. Identify possible causes and suggest improvements based on image formation models.
13. Explain how the features of Computer Vision systems enable applications such as automated surveillance.
14. A system processes satellite images for land analysis. Explain how digital image fundamentals and acquisition methods influence the accuracy of results.
15. Given an application like OCR (Optical Character Recognition), explain how sampling, quantization, and image acquisition collectively affect performance.
16. A high-resolution image consumes excessive storage and processing time. Explain how sampling and quantization can be adjusted to optimize performance without significant quality loss.
17. An image captured under uneven lighting shows poor contrast. Explain how the image formation process contributes to this issue and suggest corrective measures.
18. In a medical imaging system, explain how improper quantization can lead to loss of critical diagnostic information.
19. A self-driving car system relies on visual input. Explain how image acquisition and Computer Vision features contribute to reliable decision making.
20. An application requires detecting fine details in images. Explain how sampling rate and intensity resolution should be chosen to meet this requirement.
21. An image captured at a low sampling rate shows jagged edges. Explain how sampling rate affects spatial resolution and suggest improvements.
22. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.
23. In a thermal imaging system, explain how image sensing affects temperature-based visualization accuracy.
24. A blurred image is obtained due to motion during acquisition. Explain how image formation and sensing contribute to this effect.
25. A satellite image shows loss of fine terrain details. Explain how sampling and resolution affect feature visibility.
26. In a biometric system, explain how image acquisition quality affects recognition accuracy.

27. An image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.
28. A digital image shows reduced contrast due to poor sensing conditions. Explain how acquisition parameters influence this.
29. In a machine vision system, explain how improper quantization affects defect detection.
30. A camera captures images with varying brightness levels. Explain how image formation models influence intensity variations.
31. An image processing system fails to detect edges clearly. Explain how sampling and acquisition affect edge visibility.
32. In an industrial inspection system, explain how image sensing impacts defect detection reliability.
33. A low-bit image loses important details. Explain how quantization affects information preservation.
34. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.
35. A digital image suffers from aliasing when resized. Explain how sampling principles apply in this scenario.
36. In medical imaging, explain how increasing sampling rate improves diagnostic accuracy.
37. A system captures images with inconsistent quality. Explain how acquisition methods influence uniformity.
38. A high-speed camera captures distorted images. Explain how image formation and sampling contribute to distortion.
39. In an OCR system, explain how poor sampling affects character recognition accuracy.
40. A digital image shows quantization noise. Explain its origin and suggest mitigation techniques.
41. A vision system processes images with varying illumination. Explain how image formation models help normalize such variations.
42. An image appears overly smooth with loss of detail. Explain how quantization levels contribute to this issue.
43. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

44. A camera system produces images with color distortion. Explain how sensing and acquisition affect color representation.
45. A digital image has insufficient contrast. Explain how image acquisition and formation influence contrast levels.
46. A system requires detecting small objects in images. Explain how sampling and resolution should be adjusted.
47. In satellite imaging, explain how quantization affects classification accuracy.
48. A vision system fails in low-light conditions. Explain how image sensing impacts system performance.
49. A digital image contains staircase artifacts. Explain how sampling causes this effect.
50. In an industrial system, explain how acquisition noise affects measurement accuracy.
51. A camera captures images with motion blur. Explain how image formation and acquisition timing affect clarity.
52. A digital image has poor dynamic range. Explain how quantization influences intensity representation.
53. In a real-time vision system, explain how acquisition speed impacts processing efficiency.
54. A system captures images with uneven resolution across regions. Explain how sampling contributes to this issue.
55. In medical imaging, explain how image acquisition affects diagnostic reliability.
56. A digital image appears overly compressed. Explain how sampling and quantization contribute to information loss.
57. A vision system detects objects incorrectly due to poor acquisition. Explain the relationship between sensing and detection accuracy.
58. A high-resolution image still lacks detail. Explain how image formation models influence clarity.
59. In an automated inspection system, explain how sampling affects defect detection sensitivity.
60. A camera captures images with inconsistent intensity levels. Explain how quantization and sensing contribute to this variation.

Code of Conduct:

I Bhavya PS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A handwritten signature in black ink, reading "Bhavya PS", is centered within a light gray rectangular box.

Signature of the student:

Q1] A Camera Captures an image that appears blurred due to Improper sampling. Explain how sampling effects Image quality and suggests a method to correct it.

Answer:

Sampling is the process of converting a continuous Image into discrete pixels. the sampling rate determines the spatial resolution of the digital Image. If the sampling rate is too low fewer pixels are used to represent the Image, causing loss of fine details and producing a blurred appearance.

The problem can be corrected by:

- Increasing the sampling rate using a higher resolution sensor

- Applying interpolation techniques such as bilinear or bicubic interpolation when resizing Images.

- using optical anti-aliasing filters before sampling to Preserve Image details.

Proper sampling improves Image sharpness and allows accurate feature extraction for computer vision applications.

Q2] Differentiate how pixel resolution and intensity resolution affect Image representation in a practical application like medical Imaging.

Answer:

Image quality depends on both pixel resolution and Intensity resolution.

Pixel resolution refers to the number of pixel used to represent an image. higher pixel resolution captures more spatial details

Intensity resolution refers to the number of brightness or gray levels used to represent each pixel. Higher intensity resolution provides smoother brightness variations and better tissue differentiation.

In medical imaging:

→ high pixel resolution helps detect tiny tumors, fractures and blood vessels.

→ high intensity resolution distinguishes between tissues having similar brightness values.

Q3] A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Answer:

Quantization is the process of converting continuous grayscale intensity values into a finite number of digital levels. Each pixel is assigned the nearest available intensity value.

For example, an 8-bit image uses 256 gray levels (0-255).

If more quantization levels are used:

- more image details are preserved.
- Smooth intensity transitions are maintained.
- Better contrast is obtained.

If fewer quantization levels are used:

- Image details are lost.
- Banding effect appears.
- Diagnostic accuracy decreases.

Q4] Explain how image formation models help in improving image quality in application such as autonomous driving.

Answer:

Image formation models explain how light from object passes through the camera lens and forms an image on the sensor. The quality of the captured image depends on illumination, camera position, lens properties, and sensor characteristics.

- correcting lens distortion
- compensating for varying lighting conditions
- Improving image focus.
- Enhancing object visibility during rain, fog or nighttime.

Q5] Explain how the features of computer vision systems enable applications such as automated surveillance.

Answer:

The main features include

- Image acquisition using cameras
- Image preprocessing to remove noise
- object detection and tracking.
- face and vehicle recognition.
- motion analysis
- Automatic decision-making using AI

Thus computer vision improves surveillance efficiency, accuracy, and security.

Q6] Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical application.

CCD Sensor	CMOS Sensor
higher image quality	good image quality
lower noise	slightly higher noise
higher power consumption	lower power consumption
Slower readout	faster readout
more expensive	less expensive.

Application

→ CCD sensors are preferred in medical imaging, astronomy, and scientific research where image quality is critical.

→ CMOS sensors are widely used in smartphones, webcams, surveillance cameras, and autonomous vehicles because they are faster and more energy efficient.

Q7] In a medical Imaging system, explain how improper quantization can lead to loss of diagnostic information.

Answer:

If Improper quantization uses too few gray levels:

- Different tissues appears similar
- small tumours and lesions may disappear
- Important diagnostics details are lost
- the probability of incorrect diagnosis increases.

Therefore, selecting appropriate quantization levels is essential in medical imaging.

Q9] A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

Answer:

Banding occurs when an image has too few quantization levels, causing smooth brightness changes to appear as visible bands. This reduces image quality and loses fine intensity details.

→ Increase the bit depth (e.g. 8-bit or higher)

→ use more quantization levels to achieve smoother intensity transitions.

→ Apply dithering techniques if needed

Q9] A digital image appears pixelated when zoomed. Analyze the role of sampling and resolution in this effect.

Answer:

Pixelation occurs because the image has a low sampling rate or low spatial resolution. When the image is enlarged, individual pixels become visible, making the image appear blocky.

→ Capture the image with a higher sampling rate (higher resolution)

→ use interpolation techniques while enlarging the image.

Higher sampling improves image clarity and reduces pixelation.

Q10] In a facial recognition system, explain how different levels of computer vision are involved from Image acquisition to decision making.

Answer:

1. Image Acquisition: the camera captures the face
2. preprocessing: the Image is enhanced by reducing noise and adjusting brightness.
3. feature extraction: Important facial features are identified.
- + Recognition and Decision: the features are compared with stored faces and the system identifies the person.